

COLDPOD · PROBLEM STATEMENT

ColdPod problem statement

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CPD-PRB-001	Version	0.4
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coldpod		
Source	docs/01-problem.md @ 12fbc7c		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); budget kept at \$300; origin of the idea (vaccine vial monitor) added
0.3	2026-09-25	Amish Chadha	Record Amish's decisions (CPD-DDR-001), outreach vaccinators first and budget \$300; partner stays open
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users and context, constraints, prior work with sources)
0.1	2026-09-24	Amish Chadha	Initial scaffold

ColdPod problem statement

Insulin and vaccines must stay between 2 and 8 °C in transport, but the last leg of the cold chain (a health worker on foot or on a motorbike, or a person with diabetes travelling in a hot place without steady power) relies on ice packs and luck: loads get too warm when the ice runs out, freeze when the ice is too cold, and nobody knows either happened because nothing records the temperature.

■ The problem

Both heat and cold spoil the load. Unopened insulin keeps its potency until the printed expiry date only when stored at about 2 to 8 °C (36 to 46 °F); it can be kept at 15 to 30 °C (59 to 86 °F) for up to 28 days, loses potency faster above that, and must be thrown away if it has frozen ([US FDA, insulin storage in emergencies](#)). Most routine vaccines have the same 2 to 8 °C window, and several (for example those with an aluminium adjuvant) are damaged by freezing.

Freezing is common, not rare. A systematic review found that vaccines were exposed to temperatures below the recommended range in about 33 % of storage studies in wealthier countries and 37 % in lower-income countries, and during shipment in 38 % and 19 % of studies respectively ([Hanson et al., Vaccine, 2017](#); an earlier review reached a similar conclusion: [Matthias et al., Vaccine, 2007](#)). Frozen water ice packs placed against vials are a well-known cause, which is why WHO now prequalifies "freeze-free" cold boxes and carriers ([WHO PQS E004/CB05.3](#)).

Passive carriers have a fixed clock. A WHO-prequalified vaccine carrier must keep its load cold for at least 15 h (short range) or 30 h (long range) at a constant +4 °C, and then it needs freshly frozen or conditioned ice packs ([WHO PQS E004/VC01 test protocol](#)). Where freezers are scarce or power is intermittent, the ice packs are the weak link.

Monitoring is separate and often missing. Good practice is a digital data logger with a buffered probe, ± 0.5 °C accuracy, logging at least every 30 min and an out-of-range alarm ([CDC Pink Book, chapter 5](#)). WHO's 30-day temperature recorders alarm at -0.5 °C or lower for 60 min and at $+8$ °C or higher for 10 h ([TechNet-21, 30DTR](#)). In a carrier, these are extra devices that are easy to leave behind, and they record a problem rather than prevent it.

Existing active coolers are closed or costly. Battery-powered cold chain boxes with built-in monitoring exist, for example the Ember Cube shipping box ([Fast Company](#)), and small USB-powered insulin coolers are sold to travellers ([4AllFamily Voyager](#)). None is an open, inspectable design that a clinic, university group or maker space can build, repair and adapt.

Indicators record heat, not cold. The vaccine vial monitor, a heat-sensitive label that WHO and PATH brought into use on oral polio vaccine in 1996 ([PATH](#)), showed how much a cheap indicator travelling with each vial can do, but it does "not measure exposure to freezing temperatures" ([OpenLearn Create, Immunization module](#)). That gap between heat monitoring and freeze protection is the starting point for ColdPod.

The gap ColdPod addresses: an open, repairable carrier that combines passive storage (so it works with no power), active cooling (so it can be topped up from a vehicle, a solar panel or a USB-C charger instead of a freezer) and an integrated logger that raises alarms before a load is lost.

■ Users and context

User	Need	Context
Outreach vaccinator or community health worker	Carry a day's vaccines to villages and back without freezing or overheating them; know that the load stayed in range	On foot, bicycle or motorbike; 25 to 43 °C; clinic power intermittent; 8 to 12 h trips
Person with diabetes	Keep a month's insulin cold while travelling or during power cuts	Hot climates, long journeys, displacement, homes without a refrigerator
Pharmacy or clinic running home deliveries	Deliver temperature-sensitive medicines with a record of the trip	Van or motorbike with a 12 V socket
Humanitarian or disaster response team	Carry insulin and vaccines where the grid is down	Solar panels, vehicles, generator hours
Open hardware and global health researchers	A reproducible, auditable reference design	University labs, maker spaces

■ Constraints

- Garage-buildable prototype, \$300 USD or less in parts (raised from \$250 by Amish, 2026-09-25, CPD-DDR-001 D10; kept at \$300 by CPD-DDR-002, although the TRL 3 BOM is now \$303), using off-the-shelf modules, 3D-printed or hand-made parts, and bought vacuum-insulated panels.
- Carried by one person, by hand or on a shoulder strap.
- Chargeable from a vehicle 12 V socket, a small solar panel or a USB-C Power Delivery (PD) charger; no mains-voltage parts inside the box.
- Battery small enough to carry on a passenger aircraft without airline approval (100 Wh or less per the [FAA PackSafe rules](#)).
- Research and educational prototype only. ColdPod is not a medical device, is not WHO-prequalified, and must not be the only protection for real vaccines or insulin.

■ Out of scope

- Frozen and ultra-cold products (for example −20 °C or −70 °C vaccines).
- Clinic refrigerators and bulk storage.
- Regulatory prequalification, which would follow only after lab and field validation well beyond TRL 3.
- Cloud services. Data stays on the device and the owner's phone.

■ Open questions

- Which user comes first? Decided by Amish, 2026-09-25 (CPD-DDR-001 D11): the outreach vaccinator first, since that user sets the 43 °C design case; the traveller with insulin second.
- Which partner should help shape the requirements (an immunization program, a diabetes association, or a humanitarian logistics group)? Proposed, awaiting Amish; partners are to be picked per area later.

- How hot does it really get inside a carrier strapped to a motorbike in direct sun? The 43 °C WHO test point is used until field data exists.